

99.500 Applied Mathematics for Engineering

Sample Lab Test Question 1

March 2, 2026

1. Solve the following ODE numerically and present the results in three plots:

(a) z vs t ;

(b) $\frac{dz}{dt}$ vs t ;

(c) $\frac{dz}{dt}$ vs z

$$\frac{d^2 z}{dt^2} = c_1 e^{-bz} - \text{sign}(z') c_2 \left[c_3 |z'|^{1/2} + 0.5407 e^{-bz/2} |z'|^2 \right] - g \quad (1)$$

where

$$c_1 = 26.525$$

$$c_2 = 0.0619$$

$$c_3 = 3.343 \times 10^{-3} \quad (2)$$

$$b = 11.58 \times 10^{-5}$$

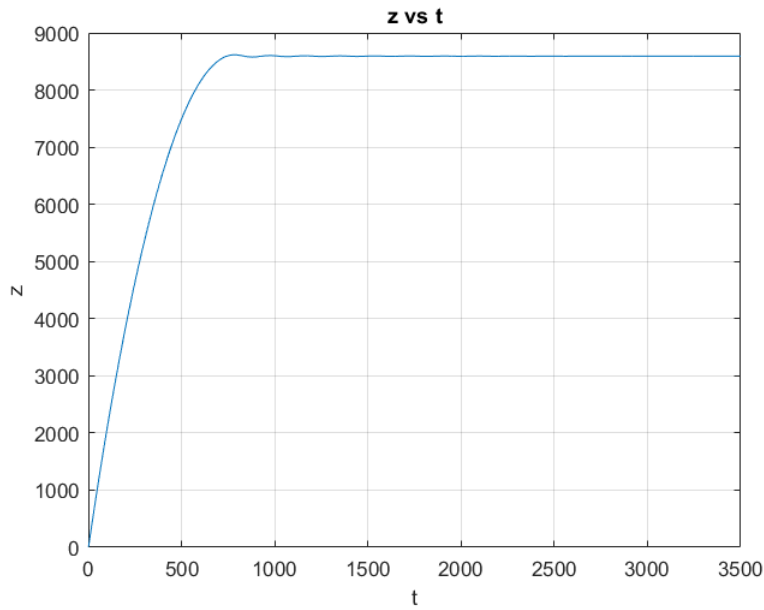
$$g = 9.807$$

with the boundary conditins

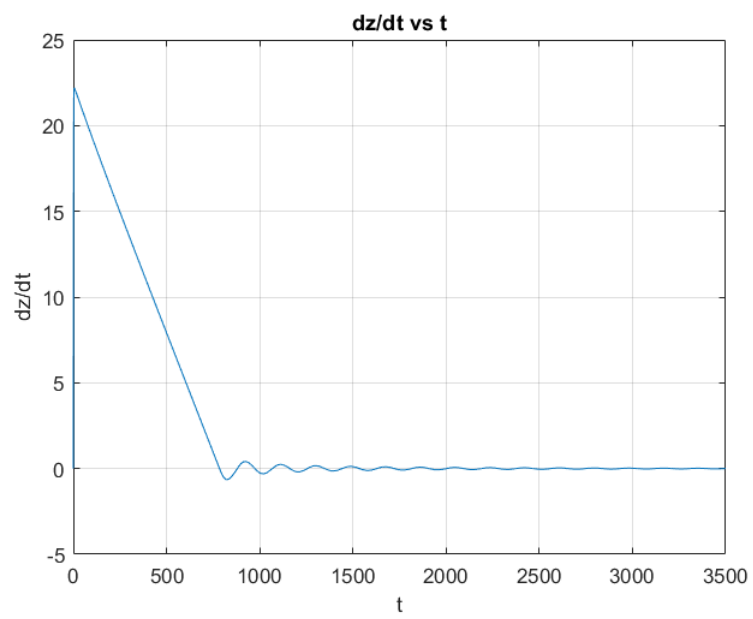
$$z(0) = 0 \quad \& \quad z'(0) = 0 \quad (3)$$

Solutions:

(a)



(b)



(c)

